

ITSS SOFTWARE DEVELOPMENT/ SOFTWARE DESIGN AND CONSTRUCTION

Lab 05: Use Case Analysis

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1. SUBMISSION GUIDELINE

To submit this lab, you and your teammate have to push all the work to the group GitHub repository, following the naming convention `MSTeamsName-TeamNumber` (e.g., TKXDPM.VP.20251-01).

For this lab, you are required to check out a branch named `release/lab05`, contains all tasks related to communication diagrams and analysis classes.

The deadline for submission will be updated by the lecturer or the teaching assistant. Therefore, you need to pay attention to avoid late submissions.

2. OVERVIEW

As previously discussed, during the architecture design process, you will work with interaction diagrams, of which the two most common types are sequence diagrams and communication diagrams. In the previous lab, you learned how to find analysis classes from the flow of events in a use case specification, understood how messages are exchanged between the instances of those analysis classes, and created completed sequence diagrams for your responsible use cases.

In this lab, you will first be guided through creating a communication diagram in **Section 3**, based on what you did with the sequence diagram in the previous lab.

Next, in **Section 4**, you will learn how to create an analysis class diagram from the interaction diagrams you have drawn in both the previous and current labs.

Finally, you will consolidate everything into a single analysis class diagram by the instruction in **Section 5**. This consolidated diagram is the output of the architecture design process and serves as a critical input for the detailed design process in the software development process, which will be continued in the next lab.

All the images/diagrams in this lab are merely examples, so *they can be inaccurate to the problem statement*. You need to check carefully to ensure that your diagram(s) are consistent with your use case model.

3. COMMUNICATION DIAGRAM

In terms of meaning, both sequence diagrams and communication diagrams are used to represent the exchange of messages between instances of analysis classes. To create a communication diagram, you also need to go through the same steps as you would for a sequence diagram.

Step 1. Find all analysis classes from flow of events in use case specification (you can skip this step, as this has been done in the previous lab).

Step 2. Create the new communication diagram. Drag all the classes and related actor(s) from the tree and drop it on the newly created diagram.

Step 3. Allocating responsibilities to classes.

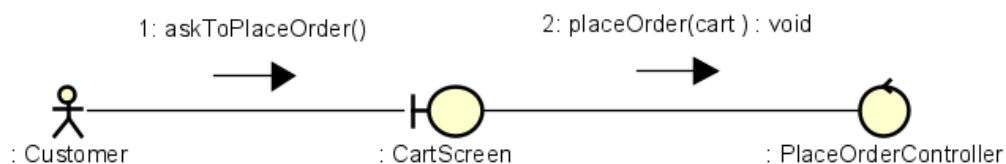


Figure 1. Simple communication diagram for use case "Place Order"

Compare with the sequence diagram you created in the previous lab, can you recognize some differences between these kinds of diagrams?

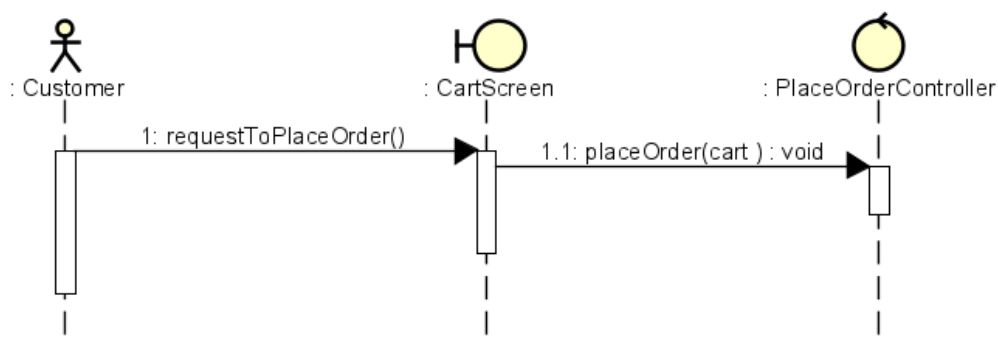


Figure 2. Simple sequence diagram for use case "Place Order"

At first glance, the communication diagram may look simpler than the sequence diagram. Depending on the purpose, you can decide which type of diagram to use. For example, during a presentation, since the slide space is quite limited, the simplicity of the communication diagram might be preferred.

Now, you will draw communication diagrams for the use cases assigned to you. You are required to create a directory named

ArchitecturalDesign/CommunicationDiagram/

to submit your work. You must submit both the Astah project file (.asta) and the exported image file (in PNG or JPG format). Don't forget to name the files according to the corresponding use case.

```
ArchitecturalDesign/
└─ CommunicationDiagram/
    └─ astah/
        └─ PlaceOrder.asta
        └─ ... (put your files here)
    └─ images/
        └─ PlaceOrder.png
        └─ ... (put your files here)
```

4. ANALYSIS CLASS DIAGRAM

From either the communication diagram or the sequence diagram, we can achieve the analysis class diagram. In this section, we will guide you through creating an analysis class diagram from a sequence diagram (the process is essentially the same for a communication diagram), by taking the use case “Place Order” as an example.

Step 1. In Astah UML, create the new class diagram **in the same project/file** with the created sequence diagram (or communication diagram).

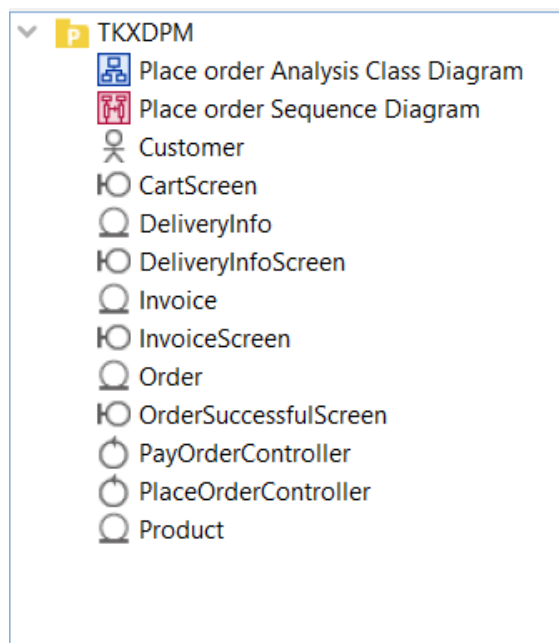


Figure 3. Create a new analysis class diagram

Step 2. In the created sequence diagram (or communication diagram), click on the message name. Then, in the box located at the bottom-left corner, click the “New” button (see Figure 4).

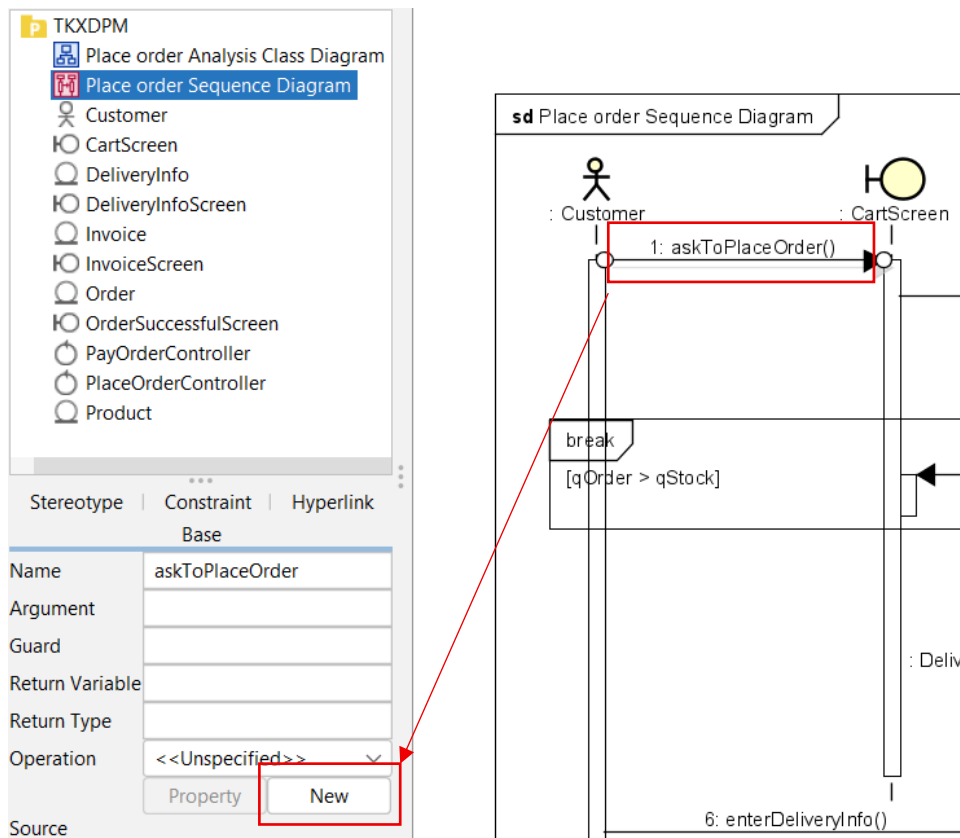


Figure 4. Click on the message name

You will see the dialog named “operation” as shown in Figure 5:

Figure 5. Convert the messages to operations

At this step, you are converting the **messages** exchanged between instances of the analysis classes in interaction diagram to the real **operations** of the class. Define each operation's signature by filling in all the required information in the dialog (e.g., return type, visibility, static, abstract, etc.). Once finished, click the "Close" button to save your changes. **Repeat this process for every message in your sequence diagram (or communication diagram).**

After this step, all analysis class diagrams will be completed with the necessary operations, as shown in Figure 6 (which are derived from the "message passing" between their instances in the interaction diagrams).

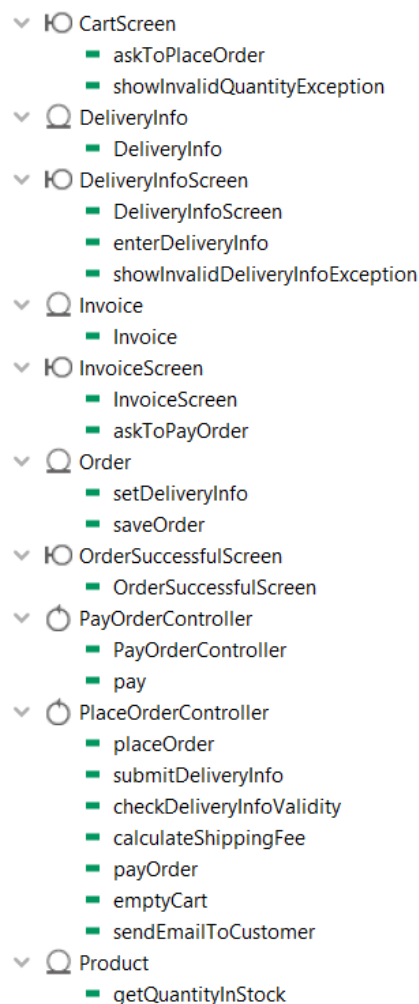


Figure 6. Analysis class with operations

Step 3. Drag and drop all the analysis classes into the analysis class diagram you created in Step 1. At first, it may look a bit messy, so take some time to rearrange the elements to make the diagram clear and easy to read, as illustrated in Figure 8.

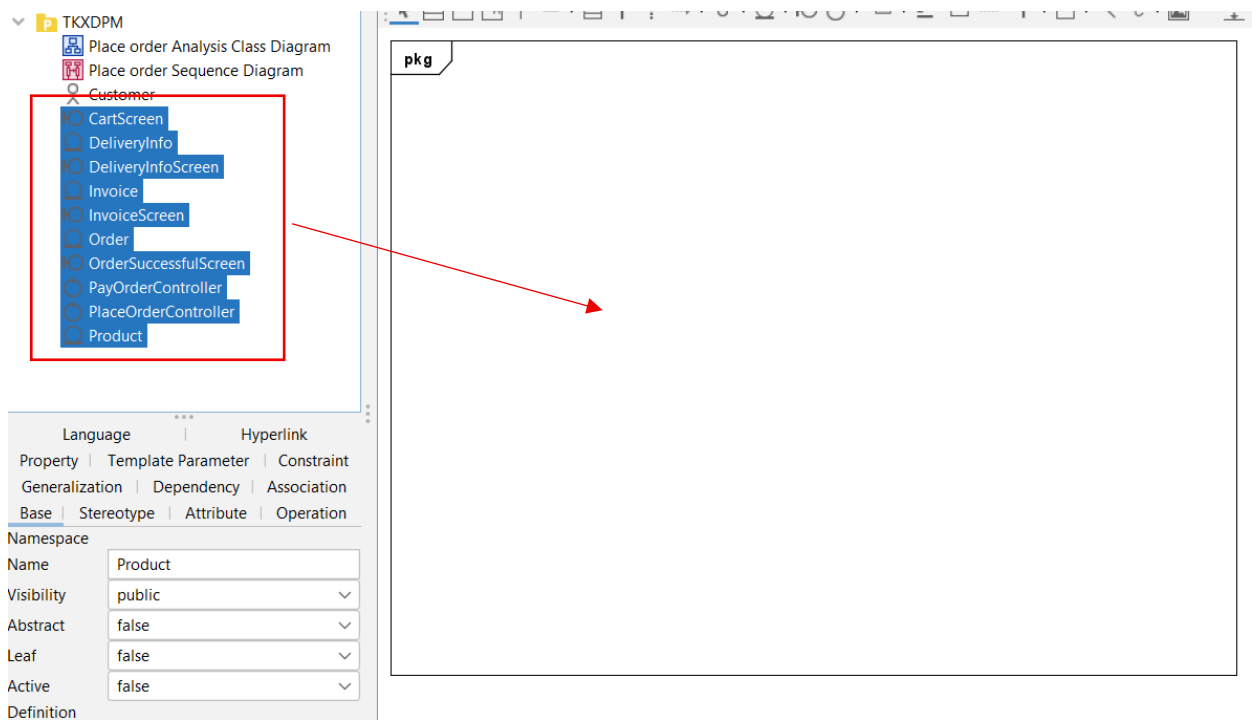


Figure 7. Drag all analysis classes to the diagram

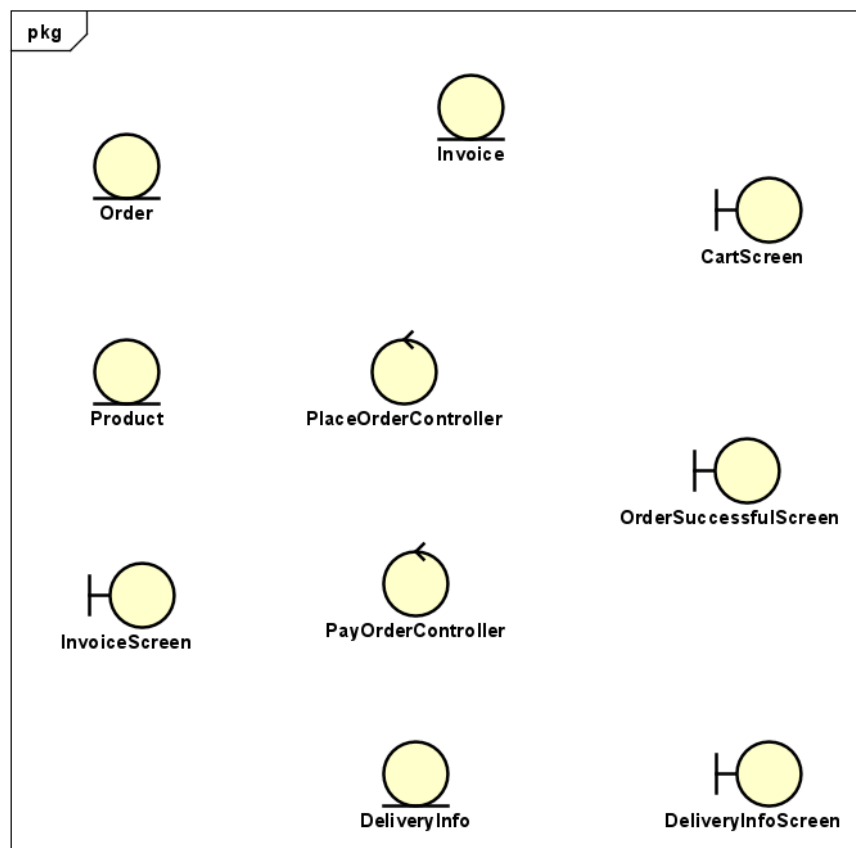


Figure 8. The initial analysis class diagram

Now, select all the classes in the current class diagram, right-click, choose “Icon Notation”, and then select “Normal” (as shown in Figure 9) to obtain the result as shown in Figure 10.

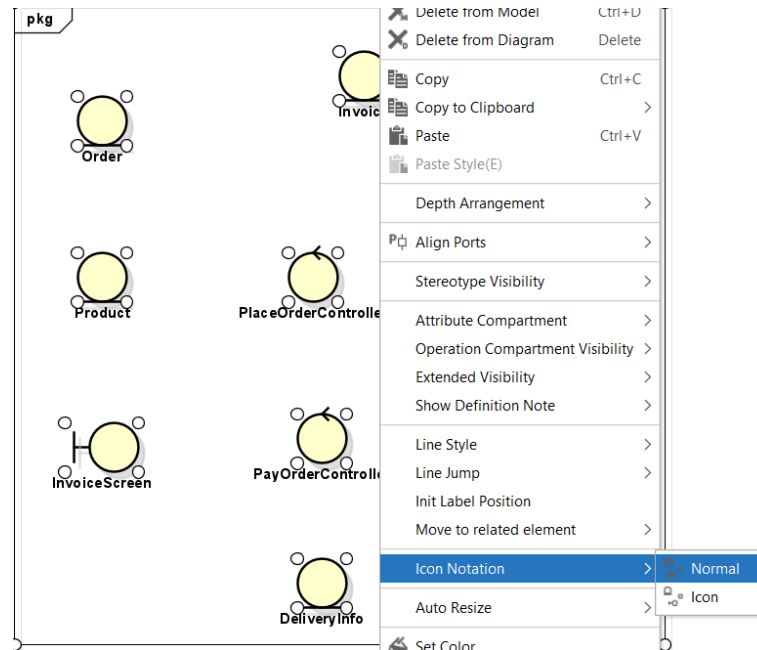


Figure 9. Convert "Icon" to "Normal" notation

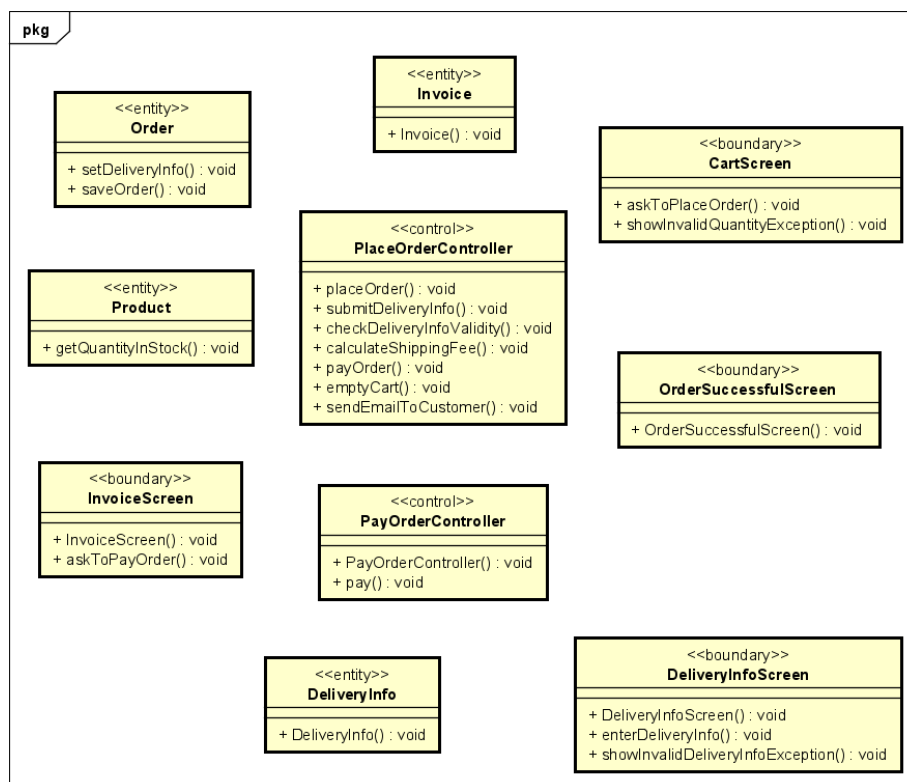


Figure 10. The analysis class diagram

Step 4. Complete the analysis class diagram by defining the relationships between classes and adding the necessary attributes for each class.

Since the analysis class diagram is only a “draft” and serves as the input for the next step – detailed design process, all relationships between classes are, by default, treated as associations. In the next lab, you will redefine these relationships more precisely. Figure 11 is the completed analysis class diagram for use case “Place Order”.

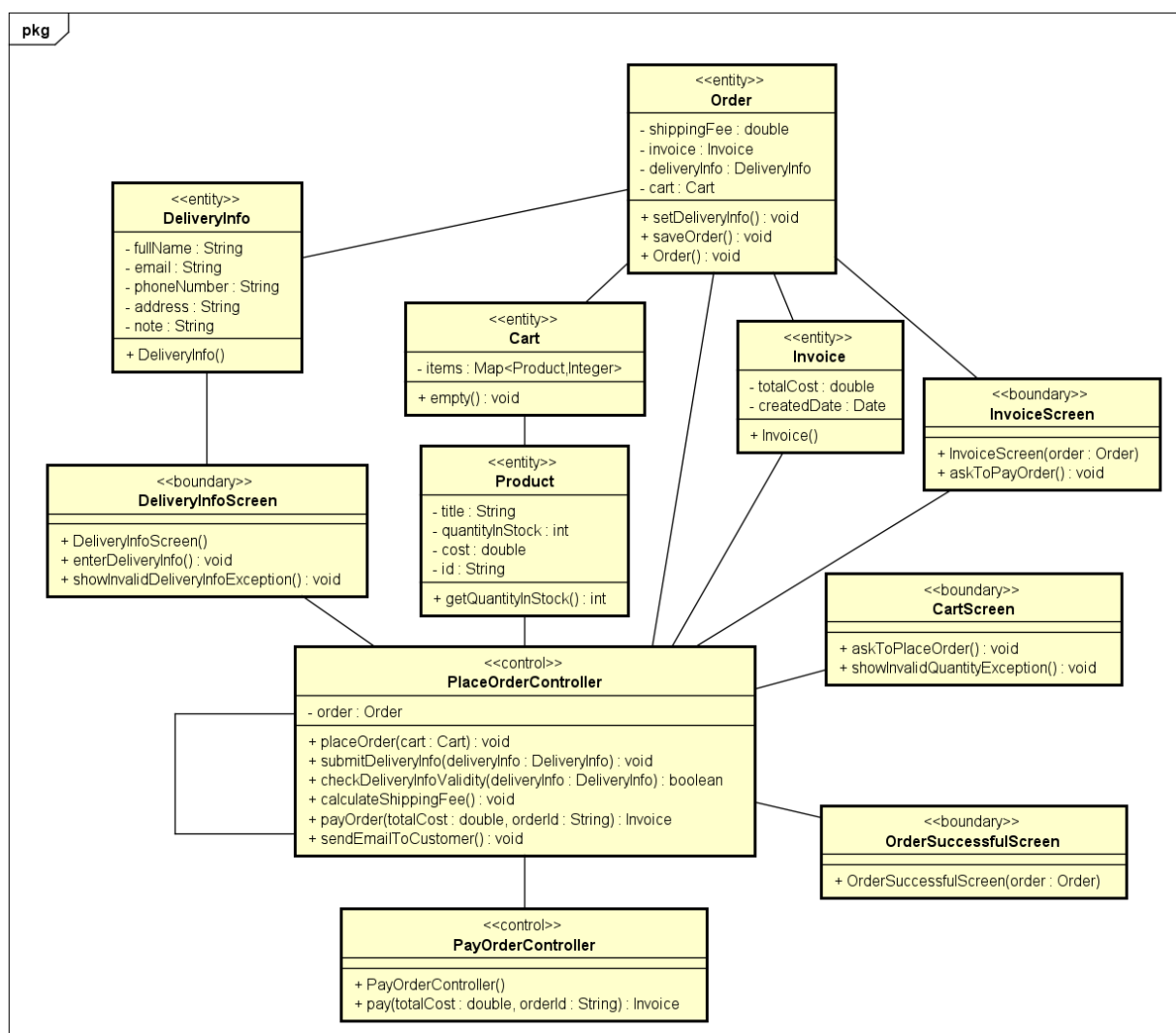


Figure 11. Completed analysis class diagram for use case "Place Order"

Actually, if you choose to create the analysis class diagram from a communication diagram, you can safely skip this step, since the relationships between classes have already been defined in the communication diagram.

Now, you will draw analysis class diagrams for the use cases assigned to you. You are required to create a directory named

ArchitecturalDesign/AnalysisClassDiagram/IndividualUC

to submit your work. Since you created the analysis class diagram in the same ASTAH file as the sequence (or communication) diagram, for this part you only need to submit the images (in PNG or JPG format) of all the analysis class diagrams, without the ASTAH file itself. Don't forget to name the files according to the corresponding use case.

```
ArchitecturalDesign/
└─ AnalysisClassDiagram/
    └─ IndividualUC
        └─ PlaceOrder.png
        └─ ... (put your files here)
```

5. MERGED ANALYSIS CLASS DIAGRAM

After completing the analysis class diagrams for each individual use case, you will combine them into a single, unified analysis class diagram. This serves as the final output of the architectural design process in the software development process.

Step 1. Create a new Astah file named **CombinedAnalysisClassDiagram**, then import all the Astah files you previously used to build the analysis class diagrams for each individual use case into this file.

In Astah UML, there is a feature that can help you with this task, the **Merge Project** feature (fully available in the free student edition) as shown in Figure 12.



Figure 12. Use "Merge Project" feature in Astah UML

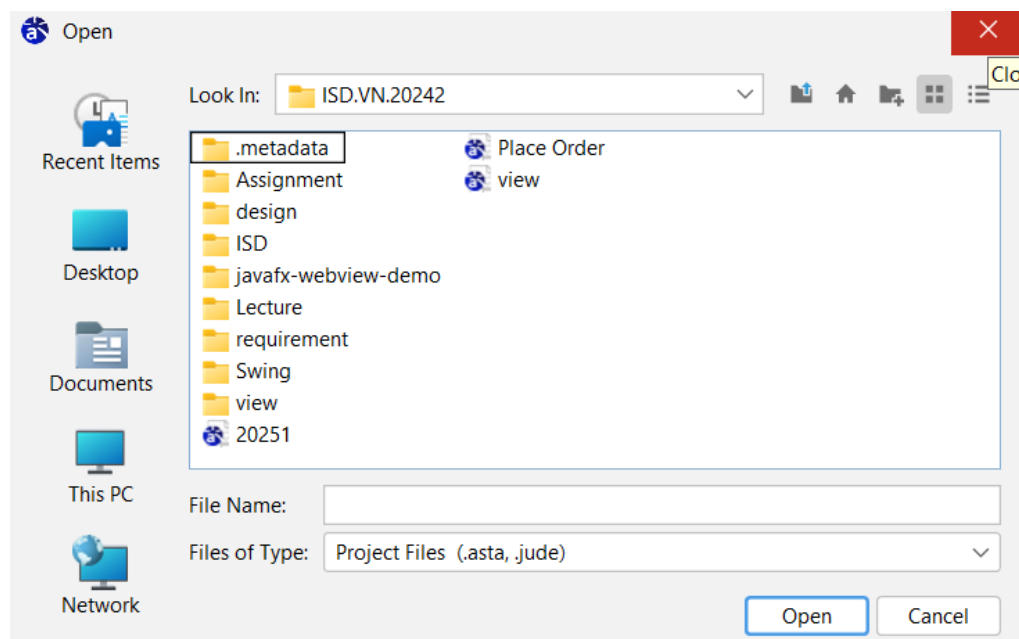


Figure 13. Choose the files you want to merge

⚠ Here are something that we want to warn you related to this step:

- When using this feature, conflicts can arise if team members are on different Astah UML versions. Align with your team and standardize on a single version for the whole project before merging.
- Team members must agree on consistent naming for analysis classes and their operations, especially if they are reused in multiple places. Otherwise, Astah UML will treat them as separate classes/operations, even if they represent the same concept. For example, *Product* and *Products* will become two different classes after merging due to the naming difference.

Step 2. After merging all the analysis classes for each use case into a single file, you can simply follow the steps outlined in **Section 4** to complete the analysis class diagram for all use cases.

Now, you will create this combined analysis class diagram. You are required to create a directory named

ArchitecturalDesign/AnalysisClassDiagram/Combined

to submit your work. You must submit both the Astah project file (.asta) and the exported image file (in PNG or JPG format). Don't forget to name the files according to the corresponding use case.

ArchitecturalDesign/
 └─ AnalysisClassDiagram/
 └─ Combined
 └─ CombinedAnalysisClassDiagram.asta
 └─ CombinedAnalysisClassDiagram.png

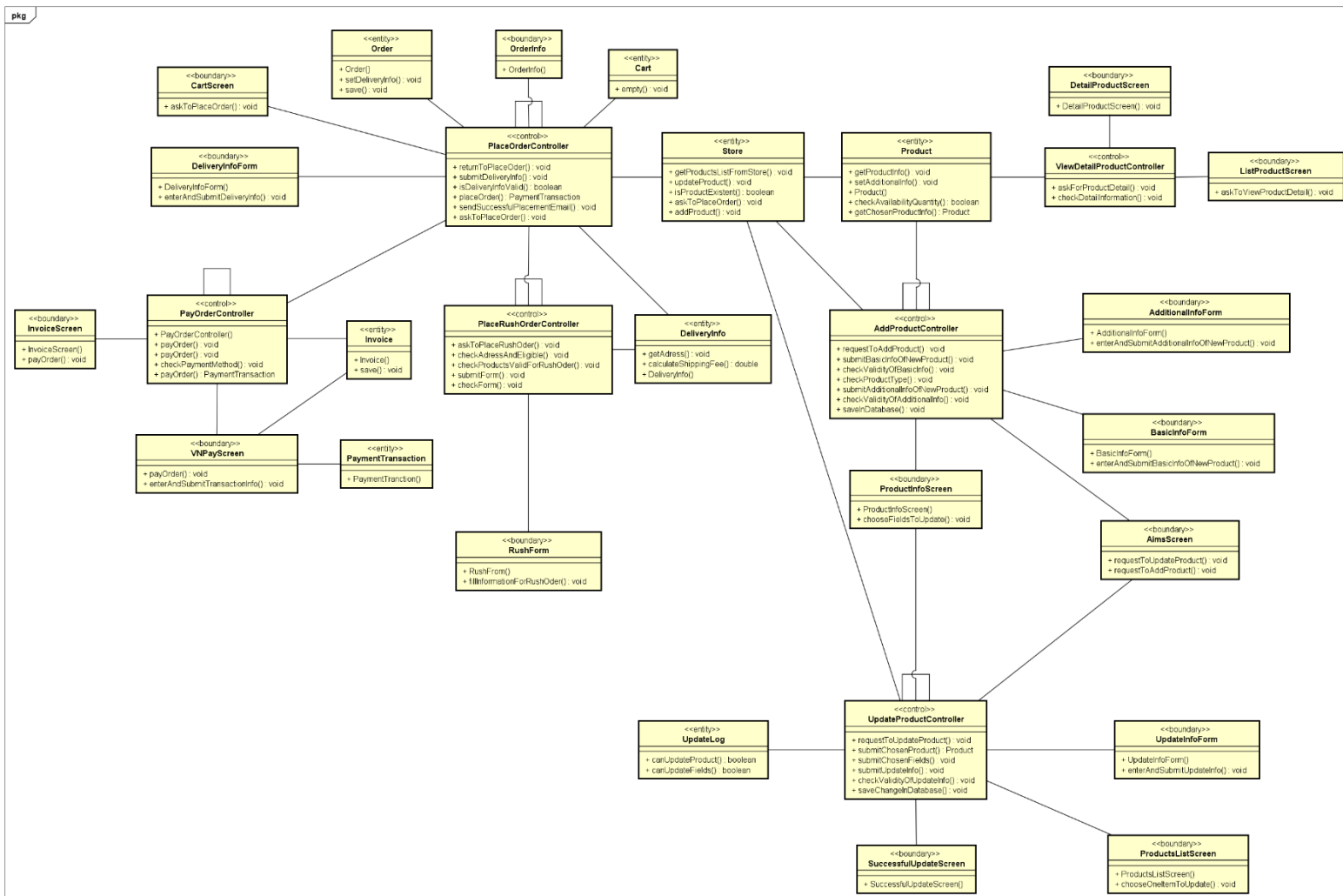


Figure 14. A sample for combined analysis class diagram